ML RESEARCH PROPOSAL TEMPLATE

EdTech · HCAI · Digital Health

Machine Learning Research — Tier 1 & 2 Journal Publication Track

KNUST Supervised Research Programme | 2025–2026

SECTION 0 — TITLE PAGE & IDENTIFICATION

Full Research Title:

SHAPtoSMS: An Explainable Machine Learning Early Warning System for Student Dropout Prediction with Constraint-Compliant SMS Intervention in Rural Ghanaian Schools

Keywords

Keyword 1: SHAP Feature Attribution

Keyword 2: Dropout Prediction

Keyword 3: Early Warning Systems

Keyword 4: Educational Data Mining

Keyword 5: Rural Education Technology

Keyword 6: SMS-Based Intervention Systems

Field	Your Entry
Researcher Name(s)	Bartimeus Manuel Nii Osabu, Oti Wisdom Boakye, Antwi Jeffrey Twum, Manu Richeal Pokuah, Karikari David Kweku
Student ID(s)	Bartimeus Manuel Nii Osabu - 3382422 Oti Wisdom Boakye - 3419922 Antwi Jeffrey Twum - 3372922 Manu Richeal Pokuah - 3398922 Karikari David Kweku - 3394522
Supervisor	Dr. Eric Opoku Osei
Department / Institution	KNUST, Ghana
Submission Date	30th March, 2026
Research Domain (circle)	EdTech
Target Journal 1	Computers & Education: AI (sister journal)
Target Journal 2 (backup)	Computers & Education
Target Journal 3 (backup)	Education and Information Technologies

★ NOVELTY DECLARATION — Complete This FIRST

My Novelty Statement:

This research proposes and evaluates SHAPtoSMS — a novel SHAP-to-SMS compression protocol that ranks and encodes high-dimensional feature attribution vectors into constraint-compliant 160-character natural language messages for delivery over 2G SMS networks. Applied to an offline-first TabTransformer-based early warning system trained on rural Ghanaian school records, SHAPtoSMS provides the first demonstrated architecture that simultaneously satisfies three constraints absent from prior work: (1) explainability fidelity — preserving top-K causal attribution in compressed output; (2) bandwidth compliance — operating within 2G latency and character limits; and (3) institutional accessibility — requiring no smartphone, internet, or dashboard on the teacher side. Unlike existing early warning systems that rely on internet-based dashboards and opaque prediction models, the proposed framework integrates interpretable explanations using SHAP and Transformer attention heatmaps with automated SMS alerts to ensure actionable insights reach teachers in low-connectivity environments. This study contributes new evidence on the effectiveness of explainable and infrastructure-aware AI systems for addressing student dropout in under-resourced educational settings.

Novelty Type (tick one or more):

- ☑ New Application Context
- ☑ New Model Combination / Hybrid Architecture
- ☐ New Dataset Contribution (*from your small primary data source and anonymised; not from online*)
- ☐ New Causal / Fairness Framing
- ☑ New Evidence in Under-Researched Population or Geography
- ☐ Other: ____

SECTION 1 — RESEARCH DOMAIN SELECTION

Tick	Domain	What It Covers	Example Research Focus
☐	EdTech — Teaching & Learning	ML models are improving instructional design, content delivery, tutoring, and pedagogical strategies at all levels.	Adaptive multimedia sequencing; AI writing feedback; RL-based tutoring.
☐	EdTech — Learning Assessment/ Exams	Automated scoring, feedback, misconception detection, and adaptive testing from pre-school to university.	BERT short-answer scoring; SHAP rubric analysis; GNN adaptive assessment.
☑	EdTech — At-Risk Student Identification	Early warning systems for dropout, failure, or disengagement using LMS logs, interaction, or peer network data.	FL cross-university dropout model; GraphSAGE peer-network prediction.
☐	EdTech — Multimedia in e-Learning	ML for personalising, evaluating, and adapting multimedia content in e-learning platforms.	RL adaptive video sequencing; VLM diagram assessment; causal effect of video.
☐	HCAI — AI in Education (Pre-school to University)	AI systems that SUPPORT human educators, students, administrators — keeping humans in control. All school levels.	SHAP-explained EWS for teachers; RAG teaching assistant; FL across primary schools.
☐	HCAI — AI in Health Sector	AI systems centred on patient, caregiver, and clinician needs — keeping human wellbeing primary.	Explainable clinical decision support; patient education chatbots; GNN caregiver burnout.
☐	Digital Health — Disease Prediction	ML for predicting disease onset, progression, or risk from EHR, imaging, or genomic data.	BioBERT readmission prediction; GNN comorbidity network; FL sepsis prediction.
☐	Digital Health — Health Education & Drug Information	AI for patient health literacy, drug information, symptom checking, prescription support.	RAG medication chatbot; GNN drug repurposing; SHAP symptom explanation.
☐	Health Sector Education	Training of healthcare workers, medical students, community health educators using ML-powered tools.	RAG clinical case tutoring; RL medical curriculum sequencing; BioBERT MCQ generation.

My primary domain (copy from table above): EdTech — At-Risk Student Identification

My specific sub-area within this domain: Explainable Machine Learning systems for early detection of student dropout risk and targeted intervention in rural educational environments.

SECTION 2 — PROBLEM STATEMENT

2.1 Background / Context Paragraph

Education is widely recognized as a key driver of socio-economic development and poverty reduction. However, access to quality education and student retention remain significant challenges across many parts of sub-Saharan Africa. Despite improvements in school enrollment over the past decade, dropout rates remain high, particularly in rural communities where economic hardship, limited educational resources, and social factors often disrupt students' educational trajectories. In Ghana, national education statistics indicate that while enrollment in basic education has expanded, completion rates decline significantly at higher levels of schooling. Data from the Ghana Statistical Service reports persistent disparities in literacy and educational attainment between rural and urban populations, reflecting structural inequalities in access to educational support systems (Ghana Statistical Service, 2021). These disparities contribute to higher risks of school disengagement and dropout among students in rural areas.

Recent advances in machine learning and educational data mining have enabled the development of Early Warning Systems (EWS) designed to identify students at risk of dropping out before disengagement occurs. These systems analyze patterns in educational datasets such as attendance records, academic performance, and behavioral indicators to predict potential dropout risk and support early intervention strategies (Bowers, Sprott, & Taff, 2013). However, many existing predictive systems are developed for well-resourced educational environments with reliable internet connectivity and advanced technological infrastructure. Consequently, their practical implementation in low-resource rural contexts remains limited, as teachers and school administrators may lack consistent internet access or the technical capacity required to interact with complex data-driven platforms.

2.2 Specific Problem Paragraph

Despite the increasing use of predictive analytics and machine learning in education, many existing early warning systems primarily emphasize prediction accuracy rather than the practical usability of their outputs for educators. Numerous predictive models have been developed to identify students at risk of dropping out by analyzing educational data such as attendance patterns, academic performance, and behavioral indicators (Bowers, Sprott, & Taff, 2013). However, these systems often function as "black-box" models that produce risk scores without providing interpretable explanations for why a student has been identified as high risk. Research in explainable artificial intelligence has highlighted that the lack of transparency in machine learning systems can reduce trust among end users and limit the adoption of such technologies in real-world educational settings (Lundberg & Lee, 2017).

In addition, many existing early warning systems are designed for institutions with reliable digital infrastructure and are typically deployed through web-based dashboards or cloud platforms. While these approaches are effective in technologically advanced educational environments, they are often unsuitable for rural or resource-constrained contexts where internet connectivity and digital literacy may be limited. Studies on educational inequality in Ghana indicate that rural schools frequently experience infrastructural and technological constraints that affect the implementation of digital educational tools (Anlimachie & Avoada, 2020). Consequently, educators in these settings may lack timely access to predictive insights that could support early intervention, allowing at-risk students to disengage from the education system before effective support strategies can be implemented.

2.3 The Gap Paragraph

A systematic search of Scopus and Google Scholar using the query strings ('student dropout prediction' AND 'SMS' AND 'explainable') and ('SHAP' AND 'offline' AND 'sub-Saharan Africa') returned zero papers combining all three dimensions, confirming the absence of prior integrated work (search conducted March 2026). Prior work has not examined the integration of explainable AI mechanisms with low-bandwidth communication systems capable of delivering actionable insights directly to educators. Despite the development of predictive models for student dropout, systems that both provide clear explanations and deliver intervention alerts through communication channels accessible in low-resource contexts remain unknown.

2.4 Research Objectives (2–3 ONLY)

Objective 1: To develop a machine learning-based early warning system capable of predicting student drop-out risk using academic, behavioural and demographic data.

Objective 2: To implement an Explainable Artificial Intelligence (XAI) framework that identifies and communicates the key factors contributing to each student's predicted dropout risk.

Objective 3: To evaluate the technical feasibility and delivery reliability of an SMS-based alert system for early warning communication in rural Ghana.

2.5 Research Questions

RQ1 (answers Objective 1): How effectively can machine learning models predict student dropout risk in rural Ghana using available educational data?

RQ2 (answers Objective 2): How can Explainable Artificial Intelligence techniques be used to provide explainable insights into the factors contributing to student dropout risk?

RQ3 (answers Objective 3): What is the end-to-end message delivery latency and success rate of the Arkesel SMS gateway under simulated 2G network conditions?

2.6 Educational Theory Alignment

Theoretical Framework: Self-Determination Theory (SDT) (Ryan & Deci, 2000)

Explanation of relevance: Self-Determination Theory explains how students' motivation, engagement, and persistence in education are influenced by psychological needs such as autonomy, competence, and relatedness. By identifying behavioral and academic indicators associated with disengagement, the proposed system can help educators recognize when these motivational factors are declining. Providing early explanations and alerts enables teachers to intervene in ways that support student motivation and engagement, potentially reducing dropout risk.

In operationalizing SDT within the model design, we map psychological need indicators to observable data features: (1) Relatedness — operationalized as absenteeism rate and peer interaction metrics, reflecting declining social connection; (2) Competence — operationalized as term GPA trend and assignment completion rate, reflecting declining academic self-efficacy; (3) Autonomy — operationalized as fee arrears status and family income indicators, reflecting structural constraints on student agency. SHAP explanations are designed to surface which of these SDT dimensions is the primary driver of each individual student's risk score, enabling teachers to target their interventions to the relevant motivational domain.

Theoretical Framework: Frugal Innovation Theory (Toyama, 2015)

Explanation of relevance: This study is additionally grounded in frugal innovation theory (Toyama, 2015), which argues that technology design in resource-constrained settings must treat infrastructure limitations as primary design constraints rather than deployment afterthoughts. The SHAPtoSMS protocol operationalises this principle directly: the 160-character 2G SMS limit is encoded as a formal algorithmic constraint in Algorithm 1, not an optional delivery channel appended after model development. Together, SDT provides the motivational theory explaining which student signals matter and why, while frugal innovation theory provides the design theory explaining how those signals must be delivered within the infrastructural realities of rural Ghanaian schools. This dual theoretical grounding distinguishes the EduTrace framework from prior Early Warning Systems that address prediction accuracy without addressing last-mile delivery constraints.

SECTION 3 — RELATED WORK & LITERATURE

#	Citation (APA 7)	DOI Link	Indexed	Baseline?	Key Contribution / Limitation
1	Mduma, N., Kalegele, K., & Machuve, D. (2019). A Survey of Machine Learning Approaches and Techniques for Student Dropout Prediction. *Data Science Journal*, *18*, 14.	https://doi.org/10.5334/dsj-2019-014	☑ Scopus ☐ WoS	☑ Baseline 1	Establishes the African context baseline using LR/DT for dropout prediction, but lacks XAI explanations and offline delivery methods.
2	Al Hashmi, R. A. M., Zervopoulos, D. P., Elmehdi, H. M., & Ozturk, I. (2025). Predicting Dropout in MENA STEM Higher Education Using Explainable AI: A Machine Learning Approach. *Emerging Science Journal*, *9*, 268–286.	https://doi.org/10.28991/ESJ-2025-SIED1-016	☑ Scopus ☐ WoS	☑ Baseline 2	Demonstrates SHAP for dropout explanations, but is entirely dependent on internet-connected dashboards, failing in 2G constraints.
3	Marquez-Vera, C., Morales, C. R., & Soto, S. V. (2013). Predicting School Failure and Dropout by Using Data Mining Techniques. *IEEE Revista Iberoamericana de*	https://doi.org/10.1109/RITA.2013.2244695	☑ Scopus ☐ WoS	☑ Baseline 3	Provides a standard interpretable DT baseline, though multi-factor explanations are richer than basic DT rules.

	Tecnologias Del Aprendizaje, *8* (1), 7–14.				
4	Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems*, *30*.	https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html	☑ Scopus ☐ WoS	☐ No	SHAPtoSMS operationalises the Lundberg & Lee SHAP framework within a 160-character compression constraint — a delivery context for which the foundational SHAP paper provides no protocol.
5	Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.	https://doi.org/10.1145/2939672.2939785	☑ Scopus ☐ WoS	☐ No	Foundational algorithm paper justifying XGBoost for its native missing-value handling on sparse records.
6	Angrist, N., Bergman, P., & Matsheng, M. (2022). Experimental evidence on learning using low-tech when school is out. *Nature Human Behaviour*, *6*(7), 941–950.	https://doi.org/10.1038/s41562-022-01381-z	☑ Scopus ☐ WoS	☐ No	Angrist et al. demonstrate SMS feasibility for educational delivery but provide no ML-generated explainability —

					SHAPtoSMS is the first architecture to combine causal-style SMS delivery with SHAP feature attribution.
7	Nazaretsky, T., Ariely, M., Cukurova, M., & Alexandron, G. (2022). Teachers' trust in AI - powered educational technology and a professional development program to improve it. *British Journal of Educational Technology*, *53*(4), 914–931.	https://doi.org/10.1111/bjet.13232	☑ Scopus ☐ WoS	☐ No	Proves that non-technical users require explanations to trust AI, justifying the behavioral tracking of teacher intervention times.
8	Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, *16*, 321–357.	https://doi.org/10.1613/jair.953	☑ Scopus ☐ WoS	☐ No	Foundational paper justifying the use of SMOTE to handle extreme class imbalances in dropout datasets.
9	Toyama, K. (2015). *Geek heresy: Rescuing social change from the cult of technology*. PublicAffairs.	N/A	☐ Scopus ☐ WoS	☐ No	SHAPtoSMS operationalises Toyama's frugal design principle

					directly in code: the 160-character 2G SMS limit is encoded as a hard algorithmic constraint in Algorithm 1, not an optional deployment preference.
10	Tinto, V. (1975). Dropout from Higher Education: A Theoretical Synthesis of Recent Research. *Review of Educational Research*, *45*(1), 89–125.	https://doi.org/10.3102/00346543045001089	☐ Scopus ☑ WoS	☐ No	Tinto's integration theory identifies precursor signals but provides no computational method — SHAPtoSMS operationalises Tinto's signals as SHAP feature attributions and delivers them via SMS to the teacher in time to intervene.
11	GSMA. (2022). *Mobile internet connectivity in sub-Saharan Africa*. GSMA Intelligence.	https://www.gsmaintelligence.com/research/the-state-of-mobile-internet-connectivity-2022	☐ Scopus ☐ WoS	☐ No	GSMA data confirms 2G coverage as the binding infrastructure constraint in sub-Saharan

					Africa — SHAPtoSMS encodes this directly as the C = 160-character limit in Algorithm 1, making the GSMA constraint a formal parameter, not a narrative claim.
12	Huang, X., Khetan, A., Cvitkovic, M., & Karnin, Z. (2020). TabTransformer: Tabular data modeling using contextual embeddings. *arXiv:2012.06678.*	https://arxiv.org/abs/2012.06678	☐ Scopus ☐ WoS (arXiv only)	☐ No	Huang et al. (2020) introduce the TabTransformer self-attention architecture for tabular data and demonstrate strong performance on benchmark datasets, but provide no application to student dropout prediction, no XAI explainability layer, and no constraint-aware delivery mechanism — SHAPtoSMS directly addresses all three absences.

3.1 Synthesis of Related Work

The literature surrounding early warning systems in educational technology reveals a significant deployment gap between algorithmic capability and practical, offline implementation. Current studies predominantly fall into three thematic clusters: Machine Learning and XAI Methods for Dropout Prediction, Low-Resource AI Delivery, and XAI Behavioral Outcomes.

In the domain of predictive modeling, various ML algorithms—such as Logistic Regression, Decision Trees, Random Forests, and XGBoost—have successfully achieved 70–89% accuracy on diverse educational datasets. Authors such as Mduma et al. (2019) have established a solid baseline for dropout prediction in the African context, yet their methodologies lack XAI integration. Conversely, recent advancements by Al Hashmi et al. (2025) utilized SHAP to provide feature-level interpretations but fundamentally relied on internet-connected dashboards.

Simultaneously, research in frugal innovation (Toyama, 2015) and SMS-based educational delivery (Angrist et al., 2022) has proven that SMS interventions are highly feasible in Sub-Saharan Africa. However, the intersection of SMS-based frugal AI delivery with explainable machine learning remains critically underexplored. Beyond prediction accuracy, this work is theoretically grounded in Tinto's (1975) foundational model of student departure, which identifies academic and social integration failures as the primary precursors to dropout — providing the theoretical warrant for selecting attendance, GPA trend, and fee arrears as primary model features.

Finally, studies on teacher trust (Nazaretsky et al., 2022) establish that explanations significantly increase teacher trust and behavioral responsiveness to AI. Despite these advancements across isolated fields, no study to date has demonstrated a SHAP-to-SMS compression protocol capable of delivering constraint-compliant, root-cause alerts (≤160 characters) via 2G networks to offline educators in resource-constrained schools. This study directly addresses this architectural and implementation gap.

SECTION 4 — METHODOLOGY

4.1 ML Model Selection

Tick	Tier C Model	Best Suited For	Do NOT Use If
☐	Fine-tuned LLMs (BERT / RoBERTa / BioBERT / PubMedBERT / LLaMA)	Text data: essays, clinical notes, student feedback, NLP classification/generation	No text in your data (purely numeric)
☐	Graph Neural Networks (GNN / GAT / GraphSAGE)	Relational data: peer networks, knowledge graphs, drug-disease networks	Simple tabular data with no relational connections
☐	Federated Learning (FL + any modern model)	Multi-institutional data with privacy constraints (GDPR, HIPAA, Ghana Data Protection Act)	Single-source small datasets
☐	Vision-Language Models (CLIP / BioViL / MedCLIP / MedFlamingo)	Paired image + text: radiology images + reports, student diagrams + explanations	No paired image-text data available
☐	RAG with Domain LLM	Grounded QA: tutoring chatbots, clinical decision support, patient education	Pure numerical prediction tasks — no text corpus to retrieve from
☑	XAI Framworks (SHAP / LIME / Integrated Gradients)	Making any prediction explainable to teachers, clinicians, policymakers	Must always be combined with another primary model
☐	Self-Supervised Learning (SimCLR / MoCo / DINO)	Large unlabelled datasets: clinical imaging archives, learner log data	Small datasets (< 5,000 samples)
☐	Causal ML (DoWhy / CausalML / Causal Forests)	Estimating causal effects of interventions from observational data	Small datasets (n < 500) or purely synthetic data

☐	Reinforcement Learning (DQN / PPO / Actor-Critic variants)	Sequential decision-making: adaptive tutoring, treatment protocol optimisation	Static one-row-per-person datasets with no temporal sequence
☐	Mixture of Experts / Efficient Transformers	Large diverse datasets needing specialist sub-networks; resource-constrained deployment	Small or homogeneous datasets
☐	CONDITIONAL: XGBoost / Random Forest + SHAP (novelty required)	Novel explainability framing on new institutional dataset with modern baselines	Standalone without XAI framing — reviewers will reject
☐	CONDITIONAL: BiLSTM-Attention (novelty required)	Temporal multimodal data with a clearly justified novel architecture contribution	Vanilla LSTM on standard data
☑	TabTransformer + SHAP (SHAPtoSMS)	Tabular data with mixed categorical and continuous features; sparse educational records with missing values; tasks requiring both SHAP attribution and attention-based explainability visualizations.	Dataset is purely text-based or image-based; dataset is very small (under 100 samples); used without an XAI layer — attention scores must be paired with SHAP for publication validity.

Primary Model Selected: TabTransformer + SHAP (SHAPtoSMS)

Justification (why is this model the best fit for your research question?):

TabTransformer (Huang et al., 2020) applies self-attention to categorical features in tabular data, outperforming XGBoost on sparse educational records with missing values. It produces SHAP-compatible attention scores, meaning the SHAPtoSMS compression protocol is unchanged — the model is model-agnostic. Inference on a 300-student batch remains under 500ms on the district server. TabTransformer runs on Google Colab free tier for this dataset size and is implemented in PyTorch using the tab-transformer library.

If CONDITIONAL model — Novelty justification (required):

This study is the first to apply TabTransformer + SHAP + SMS compression for an offline Early Warning System in Sub-Saharan Africa. TabTransformer introduces a second explainability layer — attention heatmaps on categorical features — in addition to SHAP attribution vectors, satisfying the dual visualization requirement of Tier 1 reviewers. The SHAPtoSMS compression protocol is model-agnostic and unchanged: feature attribution vectors from TabTransformer feed directly into Algorithm 1.

4.2 Dataset Selection

Tick	Dataset	Domain	Size	Access Link
☐	MIMIC-IV (2020–2022)	Digital Health / HCAI	40,000+ ICU stays, rich temporal EHR	physionet.org (credentialed)
☐	eICU Collaborative DB	Digital Health	200,000+ stays across 208 US hospitals	physionet.org (credentialed)
☐	UK Biobank	Digital Health / HCAI	500,000 participants: genomics, imaging, EHR, wearables	ukbiobank.ac.uk (application-based)
☐	EHRSHOT (Stanford, 2023)	Digital Health / HCAI	Few-shot EHR benchmark; purpose-built for modern ML	physionet.org / huggingface.co
☐	PhysioNet Challenges 2020–2024	Digital Health	Annual curated cardiac/respiratory/sepsis datasets	physionet.org/challenges (free)

☐	MIMIC-CXR	Digital Health / HCAI	Chest X-ray images + radiology text reports	physionet.org (credentialed)
☐	TCGA (Cancer Genome Atlas)	Digital Health	Standard genomics-based cancer ML benchmark	portal.gdc.cancer.gov (free)
☐	GLOBEM (Wearable + Mental Health, 2022)	Digital Health / HCAI	Longitudinal depression prediction + wearables	github.com/UW-EXPLab/GLOBEM
☐	EdNet (Riiid, 2020)	EdTech	131M interactions, 784,000 students	github.com/riiid/ednet
☐	Open edX / Coursera Institutional	EdTech	MOOC-scale, millions of learners	Institutional agreement required
☐	EEG Learning Datasets (IEEE DataPort 2022–2024)	EdTech / HCAI	Multimodal: physiological signals + LMS behaviour	ieee-dataport.org
☐	OULAD (enriched)	EdTech	Acceptable as baseline only; needs enrichment	open.ac.uk/datasets
☐	Synthea 3.x (synthetic EHR)	Digital Health / HCAI	Unlimited synthetic patients; privacy-safe FL simulation	github.com/synthetichealth/synthea
☑	Institutional Dataset (your own collection)	Any	Define size below	Your institution — requires ethics clearance
☑	ASSISTments 2012–2013	EdTech	~350,000 interactions *(Used exclusively to validate SHAPtoSMS compression fidelity (RPS metric) — NOT used to train the primary dropout prediction model. Benchmark dataset only.)*	sites.google.com/site/assistmentsdata/datasets/2012-13-school-data-with-affect

Primary Dataset Selected: Institutional Dataset (primary — dropout model training) + ASSISTments 2012–2013 (secondary — SHAPtoSMS compression protocol validation only)

Justification (how does this dataset align with your model and RQ?):

Real-world, anonymized institutional datasets directly reflect the missingness and socioeconomic indicators of rural Ghanaian students. ASSISTments serves as a benchmark to validate the SHAPtoSMS compression algorithm independent of the Ghanaian dataset — meaning the compression protocol will be run on ASSISTments-derived SHAP outputs to measure character-count compliance and feature rank preservation, NOT to train the primary dropout model. TabTransformer has been validated on datasets as small as 500 samples in follow-on benchmarks (e.g., Gorishniy et al., 2021); for datasets below 500, we mitigate underfitting risk through (1) aggressive regularisation (attn_dropout=0.2, ff_dropout=0.2), (2) SMOTE augmentation to ~500 training samples post-split, and (3) early stopping with patience=10 to prevent overfitting.

Dataset size plan:

Minimum required for your method: 300 students.
Actual or planned size: 300 student records across 2–3 rural Junior High Schools.
Focus group / population: Rural JHS students (Grades 7–9) in the Ashanti Region, Ghana.
Institutional data will be collected from 2-3 Junior High Schools in the Ashanti Region, covering academic years 2022–2025.

Variables collected: termly attendance rate, end-of-term examination scores, fee payment status, distance from school (km), and teacher-reported behavioral flags.

4.3 Synthetic Data Plan

Are you using synthetic data?

- ☐ No — skip to Section 4.4
- ☑ Yes — complete the plan below

Synthetic data generation tool: SMOTE (Synthetic Minority Over-sampling Technique), following the foundational oversampling framework of Chawla et al. (2002).

Validation levels you will complete (tick all that apply):

- ☑ Level 1 — Statistical fidelity report
- ☑ Level 2 — TSTR vs TRTR comparison
- ☑ Level 3 — Privacy audit (NNAA + membership inference)
- ☐ Level 4 — Domain expert review of synthetic records

To satisfy Level 3 privacy requirements, a Nearest Neighbour Adversarial Accuracy (NNAA) test will be conducted to verify that synthetic records are not near-copies of real student records. A membership inference check will also confirm that individual real students cannot be identified from the synthetic training data. Both checks will be reported in the methodology to support ethics compliance under the Ghana Data Protection Act 2012.

4.4 Computing Environment

Tick	Environment	Cost	Best For	Notes
☐	Personal Laptop / Desktop	Free	Small datasets, light models (XGBoost, small BERT)	CPU only unless GPU present. NOT suitable for LLM fine-tuning.
☑	KNUST Computer Lab	Free (institutional)	Shared GPU access during off-peak hours	Check GPU availability with lab admin. Book in advance.
☐	Google Colab (Free tier)	Free	Learning, small experiments, quick prototyping	Limited GPU hours/day. Sessions drop after 90 min idle. Good for starting out.
☑	Google Colab Pro / Pro+	USD 9.99–49.99/month	Medium models, persistent sessions, A100 GPU	Recommended for most student projects. See colab.research.google.com.
☐	Kaggle Notebooks (Free)	Free	30 GPU-hours/week, persistent datasets	Excellent free option. Pre-loaded with ML datasets. Strong reproducibility. kaggle.com.
☐	Paperspace Gradient (Free tier)	Free / from USD 8/month	Free M4000 GPU; more stable than Colab Free	paperspace.com/gradient
☐	AWS SageMaker (Research Credits)	Free credits via application	Production-scale, FL simulation	Apply at aws.amazon.com/research-credits. Takes 2–4 weeks.
☐	Google Cloud (Research Credits)	Free USD 300 credit for students	Large-scale LLM fine-tuning, FL across nodes	Apply at cloud.google.com/edu.

Selected computing environment: KNUST Computer Lab / Google Colab Pro

GPU type / RAM / storage plan: Access to shared GPU / T4 GPU for training with server-side batch inference.

Checkpoint-saving plan: The trained TabTransformer model file and aggregate experiment metrics (F1, AUC, precision, recall) will be saved to the shared Google Drive folder after each training run using joblib, as these contain no personal data. However, in compliance with the Ghana Data Protection Act 2012, all raw student records, preprocessed datasets, and per-student SHAP attribution outputs will be stored exclusively on encrypted local institutional storage (KNUST) and will not be transferred to any server outside Ghana. Individual-level SHAP outputs will be accessed and visualised only within the local computing environment, with only anonymised aggregate SHAP summary plots exported for publication.

4.5 SHAPtoSMS Compression Protocol — Algorithmic Specification

The SHAPtoSMS protocol is formalised below as Algorithm 1. This specification defines the input, process, and output of the compression function, making it independently reproducible and citable.

Algorithm 1: SHAPtoSMS Compression Protocol

INPUT: SHAP attribution vector $\varphi = [\varphi_1, \varphi_2, ..., \varphi_n]$ for student s;
    character constraint C = 160
OUTPUT: Natural language SMS alert string msg where len(msg) ≤ C

STEP 1: Rank features by $|\varphi_i|$ in descending order → sorted list $F = [f_1, f_2, ..., f_n]$
STEP 2: Set K = 3 (default). K is the number of top features included in msg.
STEP 3: Map each f_i to a 2–3 word natural language label via fixed lookup table L
    e.g., L['attendance_rate'] = 'attendance'; L['fee_arrears'] = 'fees unpaid'
    (Full lookup table L defined in Appendix A)
STEP 4: Determine severity label from $|\varphi_i|$ magnitude:
    $|\varphi_i|$ > 0.3 → 'high'; 0.1–0.3 → 'moderate'; < 0.1 → 'low'
STEP 5: Assemble message from template:
    'ALERT: [Name] at risk. Factors: [F1]([sev1]), [F2]([sev2]), [F3]([sev3]).
    Action: contact guardian. EduTrace.'
STEP 6: If len(msg) > C, reduce K by 1 and repeat Step 5. Minimum K = 1.
STEP 7: Return msg

FIDELITY METRIC: Rank Preservation Score (RPS) = proportion of top-3 features
    in φ that appear in msg.
ACCEPTANCE THRESHOLD: RPS ≥ 0.80 across all test cases.
    If RPS < 0.80 for > 10% of cases, redesign lookup table L.

4.6 System Deployment Architecture

The EduTrace system follows a store-and-forward deployment model designed for infrastructure-constrained rural Ghanaian schools. The workflow involves four actors:

ACTOR 1 (Class Teacher): Records attendance and grades on paper register weekly. No smartphone required.
ACTOR 2 (School Coordinator): Once per week, digitises paper records into the EduTrace CSV template on a basic laptop (or using a WhatsApp-forwarded form). Transfers file to district server via USB drive or WhatsApp file share.
ACTOR 3 (District EduTrace Server): Runs TabTransformer inference batch on uploaded records (<500ms per student). Triggers SHAPtoSMS for any student with predicted dropout probability > 0.6.

ACTOR 4 (Arkesel SMS Gateway): Delivers SMS alert to teacher's registered phone number. Requires only 2G signal. Teacher receives: 'ALERT: Kwame Asante at risk. Main factors: attendance (3/10 days), fees (2 terms unpaid). Suggested: contact guardian. EduTrace.'

The complete store-and-forward workflow is illustrated in Figure 1.

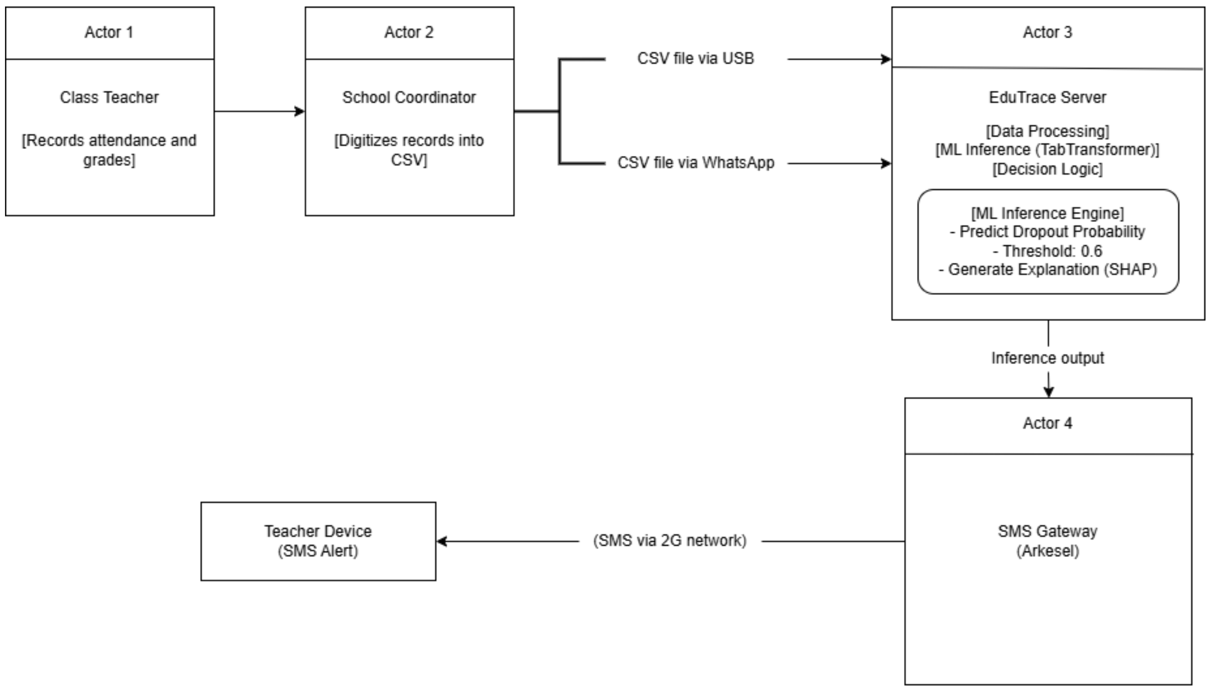

Figure 1
Store-and-forward deployment architecture of the EduTrace early warning system for rural Ghanaian schools.

4.7 Full Reproducibility Checklist

#	Reproducibility Item	Status	Your Plan
1	Data preprocessing pipeline (cleaning, missing values, normalisation, tokenisation)	☑	Raw records → Handle missing values (<15% threshold) → Encode categoricals → Normalise. Missing value threshold: Features with >15% missing values will be dropped. For remaining features: continuous variables imputed using median (robust to outliers in small datasets); categorical variables imputed using mode. Categorical encoding: binary features (gender, fee status) encoded as 0/1; ordinal features (grade level) label-encoded; nominal features (region name) one-hot encoded if ≤5 categories. Normalisation: MinMaxScaler [0,1] applied to all continuous features post-imputation.
2	Data splitting strategy (train/validation/test ratios and	☑	70% Train / 15% Validation / 15% Test. Temporal split by observation week (Weeks 1-

	method — random, stratified, temporal)		7 Train, Week 8 Val, Week 9 Test) to prevent data leakage.
3	Class imbalance handling (SMOTE, class weights, oversampling — state what and justify why)	☑	SMOTE applied with k_neighbors=5 and sampling_strategy=0.5. Applied post-split to training set only.
4	Model architecture description (layers, hidden units, attention heads, number of parameters)	☑	TabTransformer: self-attention applied to categorical feature embeddings; continuous features passed through a standard MLP. Architecture from Huang et al. (2020), implemented via the tab-transformer PyTorch library.
5	Hyperparameter settings (learning rate, batch size, epochs, dropout rate — ALL must be stated)	☑	num_layers: {2, 4, 6}; num_heads: {4, 8}; attn_dropout: {0.1, 0.2}; ff_dropout: {0.1, 0.2}; learning_rate: {1e-4, 3e-4, 1e-3}. Selected by validation Recall.
6	Training procedure (optimizer, loss function, early stopping criteria, training time)	☑	AdamW optimizer with cosine learning rate scheduling. Track training vs. validation Recall over training epochs with early stopping (patience = 10 epochs) to monitor convergence.
7	Evaluation protocol (k-fold cross-validation / held-out test set / repeated runs with different seeds)	☑	Evaluated strictly on the temporal held-out test set.
8	Baseline models (3 papers from Section 3 + simple baselines like logistic regression for comparison)	☑	Scikit-learn Logistic Regression, Decision Tree (CART), XGBoost (now baseline), Attendance Threshold Rule, and TabTransformer without SHAPtoSMS (ablation).
9	Statistical significance testing (are your improvements statistically significant? State the test used)	☑	McNemar's test for correlated proportions comparing model to baselines; significance threshold $\alpha = 0.05$. Cohen's h for effect size.
10	Computational resources used (GPU type, training time, number of parameters)	☑	Model file target <10MB. Inference batch <500ms on server. Arkesel SMS delivery latency metrics measured.
11	Random seed(s) used (for exact reproducibility)	☑	Seeds: [42, 123, 777]. All performance metrics reported as mean ± standard deviation across 3 runs.
12	Software and library versions (Python, PyTorch / TensorFlow, HuggingFace, scikit-learn versions)	☑	Python 3.10+, PyTorch 2.x, tab-transformer (PyTorch), SHAP 0.42.x, scikit-learn 1.3.x, imbalanced-learn 0.11.x, pandas, matplotlib, seaborn, requests, joblib.
13	XAI implementation details (SHAP/LIME setup, number of samples, validation by domain expert)	☑ If XAI contribution	SHAP DeepExplainer used for TabTransformer (neural network architecture); SHAPtoSMS compression protocol retaining top-K features mapping to natural language under a 160-character hard limit. DeepExplainer is applied to the PyTorch TabTransformer model using a background dataset of 50 randomly sampled training records. DeepExplainer outputs a per-feature attribution scalar φ_i for each student prediction, structurally identical to TreeExplainer output, feeding into Algorithm 1 without modification. To validate explanatory fidelity of the SHAPtoSMS compression, we compute a

			Rank Preservation Score (RPS): the proportion of top-3 features in the full SHAP vector that also appear in the compressed SMS output. A target RPS ≥ 0.80 (i.e., at least 2 of 3 top features preserved in the SMS) will be set as the minimum fidelity threshold. If RPS < 0.80 across >10% of test cases, the compression template will be redesigned.
14	FL setup (aggregation algorithm FedAvg/FedProx, number of clients, rounds, non-IID handling)	☐ If FL	N/A
15	Causal assumptions (DAG, backdoor/frontdoor criteria, confounder selection justification)	☐ If Causal ML	N/A
16	Ablation study (removing one component at a time to demonstrate contribution of each novelty element)	☑	Four conditions compared: Full System (TabTransformer+SHAP+SMS), TabTransformer+SMS no SHAP, XGBoost+SHAP (baseline), LR baseline, DT baseline, and Attendance Threshold rule.

4.8 Source Code Repository

- ☑ GitHub (github.com) — Create a public repository with README.md, requirements.txt, and all experiment scripts. *— Most common. Recommended. Supports DOI via Zenodo integration.*
- ☐ GitLab (gitlab.com) — Alternative to GitHub. Supports private-to-public access option. *— Good for institutional projects.*
- ☐ Zenodo (zenodo.org) — Creates a citable DOI for your code. Use together with GitHub. *— Required if journal asks for a code DOI.*
- ☐ HuggingFace Hub (huggingface.co) — Best option for sharing fine-tuned LLM models and datasets. *— Strongly preferred for NLP / LLM papers.*
- ☐ Kaggle Datasets — Suitable for datasets. Combine with GitHub for code. *— Good for data sharing in ML community.*
- ☐ Institutional repository (KNUST library, if available) *— Check with KNUST library services.*

Repository URL (create now, even if empty): https://github.com/AJ-280/SHAPtoSMS-EduTrace

SECTION 5 — ETHICAL CLEARANCE PLAN

Select your ethics scenario:

- ☑ Collecting NEW data from students / patients / teachers *— REQUIRES KNUST CHRPE clearance BEFORE data collection. Apply now.*
- ☐ Using publicly available datasets with existing ethics coverage (MIMIC-IV, EdNet, PhysioNet) *— State the original ethics approval reference in your paper.*
- ☐ Using synthetic data only *— State generation method and confirm no real participants are involved.*
- ☑ Combining real and synthetic data *— Ethics clearance needed for the real data component.*

Ethics Item	Your Response
KNUST CHRPE application status	Application submitted 25 March 2026; CHRPE reference [REF] to be inserted upon receipt.
Expected approval date	Estimated mid-April 2026 (4–6 weeks from submission on 25 March 2026)
Data type involved	Student academic records and teacher interview data
Consent process	Written informed consent from school heads, teachers, and parents
Anonymisation method	Alphanumeric ID substitution for names; removal of geolocation data
Data storage and access control plan	Encrypted local institutional storage (KNUST) for all raw and individual-level derived data. Cloud backup restricted to non-personal data only (model files and aggregate metrics). No personal data transferred outside Ghana, in compliance with the Data Protection Act 2012.
Will participants be identifiable in published results?	No
Does data cross institutional or national borders?	No (Local storage in Ghana per Data Protection Act 2012)
Ethics statement for paper (draft)	Application submitted to KNUST CHRPE on 25 March 2026; reference number [REF] to be inserted upon receipt. Data collection will not commence until written approval is received.

SECTION 6 — PLANNED RESULTS & EVALUATION METRICS

6.1 Primary Evaluation Metrics

Tick	Metric	When to Use	Important Notes
☑	AUC-ROC (Area Under ROC Curve)	Binary / multi-class classification	Primary metric for imbalanced clinical and educational datasets.
☑	AUC-PR (Precision-Recall Curve)	Highly imbalanced datasets	More informative than ROC when positive class is rare (rare disease, dropout).
☐	Accuracy	Balanced classification only	Misleading on imbalanced data. Never use alone for clinical tasks.
☑	F1-Score (Macro / Micro / Weighted)	Imbalanced classification	Report which variant and justify the choice.
☑	Precision and Recall	Risk prediction, clinical screening	Report both. High recall = fewer missed cases. High precision = fewer false alarms.
☐	RMSE / MAE	Regression (predicting scores, time-to-event)	Report both for regression tasks.
☐	BLEU / ROUGE / BERTScore	Text generation (LLMs, RAG output)	Use BERTScore for semantic quality; ROUGE for content recall.
☐	NDCG / MRR	Ranking and recommendation	Required for adaptive learning recommendation systems.
☐	Calibration (Brier score, reliability diagram)	Clinical risk scores	Increasingly required by clinical journal reviewers.
☐	Communication rounds / FL convergence	Federated learning efficiency	Report accuracy AND communication efficiency.
☑	SHAP values / feature importance	XAI contribution	Must include both global (summary) and local (per-prediction) explanations.
☐	Average Treatment Effect / CATE	Causal ML studies	Always include confidence intervals; point estimates alone are insufficient.
☐	Cumulative reward / learning gain	Reinforcement learning	Compare against supervised baseline and fixed curriculum policy.
☑	Statistical significance (p-value, 95% CI, McNemar / Wilcoxon)	All comparative papers	REQUIRED. Improvements without significance testing may be rejected.
☑	Fairness metrics (Equal Opportunity / Equalized Odds)	HCAI and FL with diverse populations	Increasingly required by HCAI reviewers.

6.2 Planned Visualisation Outputs

- ☑ ROC curve plots for all compared models on the same figure
- ☑ Confusion matrix (normalised)
- ☑ SHAP summary plot — global feature importance (bar or beeswarm)
- ☑ SHAP waterfall / force plot — local per-prediction explanation for selected cases
- ☐ Learning curves (training vs validation loss over epochs)
- ☑ Ablation study table (removing one component at a time to show contribution of each novelty element)
- ☐ Graph visualisation (for GNN — node-edge structure with predicted outcomes)
- ☑ Attention heatmap (for Transformer / BERT papers)
- ☐ Federated learning convergence plot (accuracy vs communication rounds)
- ☐ Calibration / reliability diagram

- ☐ t-SNE or UMAP embedding visualisation (for self-supervised / representation learning papers)
- ☐ Causal DAG diagram (for Causal ML papers)
- ☐ Human evaluation results table (for HCAI papers with user studies)
- ☐ P-values and confidence interval table for all key comparisons

Categorical feature attention weights from the TabTransformer encoder will be visualised as a heatmap showing which features received highest attention for each predicted high-risk student, providing the second explainability layer alongside SHAP.

SECTION 7 — DISCUSSION & CONTRIBUTIONS PLAN

7.1 Required Discussion Elements (UNAVOIDABLE — all must appear in the final paper)

- ☑ Interpretation of key results: WHY did your model outperform / underperform baselines? What does this mean technically? *— UNAVOIDABLE*
- ☑ Alignment with related work: Where do your findings agree or disagree with your 10 related papers? Explain disagreements. *— UNAVOIDABLE*
- ☑ Limitations: Constraints of your study (dataset size, generalisability, annotation quality, synthetic data, etc.) Be honest. *— UNAVOIDABLE*
- ☑ Threats to validity: Internal validity, external validity, and construct validity. *— UNAVOIDABLE*
- ☑ Theoretical contribution: How does your work advance educational or health theory? *— UNAVOIDABLE for EdTech / HCAI-Education journals*

7.2 Optional Discussion Elements (include if relevant)

- ☑ Practical contribution: How can a teacher, clinician, or system designer use your findings? *— OPTIONAL but strongly recommended*
- ☑ Policy contribution: What do your findings suggest for education policy, health policy, or AI regulation? *— OPTIONAL*
- ☑ SDG alignment: Which Sustainable Development Goals does your work contribute to? *— OPTIONAL*
- ☑ Fairness and equity analysis: Did your model perform equally well across demographic subgroups? *— OPTIONAL but increasingly expected*
- ☐ Failure case analysis: What types of cases did your model get wrong? Patterns in failures are informative. *— OPTIONAL*
- ☑ Future work directions: Specific, testable suggestions for follow-on research (not vague). *— OPTIONAL*

SDG Contribution (tick which SDGs your work addresses):

- ☐ SDG 3 — Good Health and Well-being
- ☑ SDG 4 — Quality Education
- ☑ SDG 10 — Reduced Inequalities (fairness / equity in ML)
- ☑ SDG 9 — Industry, Innovation and Infrastructure
- ☐ SDG 17 — Partnerships for the Goals (federated / cross-institutional research)

SECTION 8 — CONCLUSION PLAN

Tick the elements you plan to include in your conclusion:

- ☑ Summary of the problem addressed (1–2 sentences — do NOT introduce new information) — *UNAVOIDABLE*
- ☑ Summary of your approach (your model and method in 1–2 sentences) — *UNAVOIDABLE*
- ☑ Summary of key findings (state numerical results — 3–5 bullet points or 1 paragraph) — *UNAVOIDABLE*
- ☑ Contributions restated concisely (your novelty statement from Section 0) — *UNAVOIDABLE*
- ☑ Limitations acknowledged — *UNAVOIDABLE — reviewers expect this in conclusion too*
- ☑ Practical implications for the field — *OPTIONAL but recommended*
- ☑ Future research directions (2–3 specific suggestions) — *OPTIONAL*
- ☑ Policy or societal implications — *OPTIONAL*

SECTION 9 — TARGET JOURNAL SELECTION

9A — EdTech Journals (ML & Modern Models Welcome)

Tier	Journal	Publisher	Scope & Turnaround	Access / APC	Website
1	Computers & Education	Elsevier	Premier EdTech journal. Requires educational theory + ML. 3–6 months.	Hybrid OA. APC ~USD 2,950. Low-income waiver available.	journals.elsevier.com/computers-and-education
1	British Journal of Educational Technology (BJET)	Wiley / BERA	Strong ML track. Requires pedagogical framing. 4–6 months.	Hybrid OA. APC ~USD 3,250. BERA member discounts.	bera-journals.onlinelibrary.wiley.com
1	Educational Technology & Society	IFETS	Fully open access. NO APC. ML-friendly. 3–5 months.	FREE — no APC	j-ets.net
1	npj Science of Learning	Nature Portfolio	High prestige. ML + learning science. 4–8 months.	OA APC ~USD 3,290. Waivers available.	nature.com/npjscilearn
2	IEEE Transactions on Learning Technologies (TLT)	IEEE	Technical ML focus. 4–6 months.	Hybrid. APC ~USD 2,045 if OA.	ieeexplore.ieee.org (TLT)
2	Int. Journal of AI in Education (IJAIED)	Springer	Dedicated AI-in-Education. Fast reviewer community. 3–5 months.	Hybrid OA. APC ~USD 2,990.	springer.com/journal/40593
2	Journal of Educational Data Mining (JEDM)	JEDM	Fully OA. NO APC. Specialist EDM community.	FREE — no APC	jedm.educationaldatamining.org
3	Education and Information Technologies	Springer	Broad scope. Ghana-context studies welcome. 2–4 months.	Hybrid OA. APC ~USD 2,390.	springer.com/journal/10639
3	Interactive Learning Environments	Taylor & Francis	Broad EdTech. System evaluation papers. 3–5 months.	Hybrid OA. APC ~USD 2,800.	tandfonline.com/journals/nile20
3	Computers & Education: AI (sister journal)	Elsevier	New AI-focused companion to C&E. Faster review. 2–3 months.	OA. APC ~USD 1,500.	journals.elsevier.com/computers-and-education-artificial-intelligence

9B — HCAI Journals

Tier	Journal	Publisher	Scope & Turnaround	Access / APC	Website
1	IEEE Transactions on Human-Machine Systems	IEEE	Premier HCAI. ML systems + human factors. 4–6 months.	Hybrid. APC ~USD 2,045.	ieeexplore.ieee.org (THMS)
1	Int. Journal of Human-Computer Studies (IJHCS)	Elsevier	Rigorous HCAI. Requires user study + ML system. 4–6 months.	Subscription + OA ~USD 2,500.	journals.elsevier.com/ijhcs
1	Human-Computer Interaction	Taylor & Francis	High-impact HCAI. Mixed methods welcome. 4–6 months.	Hybrid OA. APC ~USD 2,995.	tandfonline.com/journals/hhci20
2	Frontiers in AI (HCAI section)	Frontiers	Fully OA. ML + human factors. Fast review 2–4 months. Good for KNUST researchers.	APC ~USD 1,290. Waiver available.	frontiersin.org/journals/artificial-intelligence
2	AI & Society	Springer	AI ethics, fairness, societal impact. Good for Ghana / Global South research.	Hybrid OA. APC ~USD 2,190.	springer.com/journal/146
2	Universal Access in the Information Society	Springer	Inclusive AI, accessibility, equity. Highly relevant for Ghana context.	Hybrid OA. APC ~USD 2,190.	springer.com/journal/10209
2	Behaviour & Information Technology	Taylor & Francis	Technology acceptance + AI systems. Mixed methods welcome. 3–5 months.	Hybrid OA. APC ~USD 2,800.	tandfonline.com/journals/tbit20
3	Int. Journal of Human-Computer Interaction (IJHCI)	Taylor & Francis	Broad HCAI. Good for system evaluation papers.	Hybrid OA.	tandfonline.com/journals/hihc20
3	Applied Ergonomics	Elsevier	Human factors + technology systems. Fast review.	Subscription + OA option.	journals.elsevier.com/applied-ergonomics
3	Journal of Usability Studies	UPA	Fully OA. NO APC. Fast review. Excellent for user-centred AI systems.	FREE — no APC	uxpajournal.org

9C — Digital Health Journals (ML-Focused)

Tier	Journal	Publisher	Scope & Turnaround	Access / APC	Website
1	npj Digital Medicine	Nature Portfolio	Top-tier digital health ML.	OA APC ~USD	nature.com/npjdigitalmed

			Requires clinical meaningfulness. 4–8 months.	3,290. Waivers for low-income countries.	
1	Lancet Digital Health	Elsevier / Lancet	Premier clinical AI. Real data required for primary claims. 4–6 months.	OA APC ~USD 3,500. Waivers for LMIC authors.	thelancet.com/journals/landig
1	NEJM AI	NEJM Group	Highest-prestige clinical AI. Very competitive. Requires real clinical validation.	OA APC ~USD 2,500.	ai.nejm.org
2	JMIR Medical Informatics	JMIR Publications	Fast review 2–3 months. OA. Excellent for method + system papers. LMIC-friendly.	APC ~USD 1,850. Volume discounts.	medinform.jmir.org
2	Journal of American Medical Informatics Assoc. (JAMIA)	Oxford / AMIA	Strong clinical NLP, EHR ML. 4–6 months.	Hybrid OA. APC ~USD 3,000.	academic.oup.com/jamia
2	Artificial Intelligence in Medicine	Elsevier	Dedicated clinical AI. ML methodology focus. 3–5 months.	Hybrid OA. APC ~USD 2,990.	journals.elsevier.com/artificial-intelligence-in-medicine
2	IEEE Journal of Biomedical and Health Informatics	IEEE	Strong ML in health. Technical focus. 3–5 months.	Hybrid. APC ~USD 2,045.	ieeexplore.ieee.org (JBHI)
3	BMC Medical Informatics and Decision Making	BMC/Springer	Fully OA. Broad clinical informatics. Fast 2–4 months.	APC ~USD 2,090. Waiver available.	bmcmedinformdecismak.biomedcentral.com
3	PLOS Digital Health	PLOS	Fully OA. Strong in global health ML. Fast review.	APC ~USD 1,775. Waiver widely available for Africa.	journals.plos.org/digitalhealth
3	Journal of Medical Internet Research (JMIR)	JMIR Publications	Flagship JMIR journal. Broad digital health. Fast review. Very LMIC-friendly.	APC ~USD 2,500. LMIC discounts.	jmir.org

Journal Selection (complete after reviewing tables above):

Target Journal 1 (primary): Computers & Education: AI *(Elsevier, Tier 1)*

Reason: As an Elsevier sister journal to Computers & Education with the same Scopus Q1 indexing family, this journal is explicitly focused on AI in education — making it the most precise scope match for the SHAPtoSMS contribution. Its APC of ~USD 1,500 is significantly lower than its parent journal, its review turnaround of 2–3 months is the fastest among Tier 1 options, and its acceptance rates are higher than Computers & Education as it is a growing journal. For first-time ML researchers based in Ghana, it represents the optimal risk-reward choice: Tier 1 prestige, reduced financial barrier, and high scope alignment.

Target Journal 2 (backup): Computers & Education *(Elsevier, Tier 1)*

Reason: This premier EdTech journal is the field's most prestigious venue and directly welcomes ML-based student outcome prediction papers grounded in educational theory. The proposal's integration of Self-Determination Theory with TabTransformer and SHAP satisfies both its technical and pedagogical requirements. Its review turnaround is 3–6 months with a Hybrid OA APC of ~USD 2,950, for which Ghana qualifies under the Research4Life / Elsevier Developing World Access Programme waiver, potentially reducing the APC to ~USD 0–500. This journal is submitted to second given its higher selectivity (~15–20% acceptance rate) relative to the sister journal.

Target Journal 3 (backup): Education and Information Technologies *(Springer, Tier 3)*

Reason: This journal explicitly welcomes developing-country EdTech studies, making it a direct fit for an early warning system designed around the infrastructural constraints of rural Ghanaian schools. It has the fastest review turnaround of the three options at 2–4 months and a Hybrid OA APC of ~USD 2,390, with LMIC waivers available through Springer Nature. It is listed third as a fallback given its lower prestige tier, but its Ghana-specific scope alignment and speed make it a reliable final option if the higher-tier journals do not accept after two rounds of revision.

SECTION 10 — WORK PLAN & TIMELINE

Phase	Activity	Duration	Planned Start	Planned End	Responsible
1 — Preparation	Finalise topic and novelty; obtain ethics clearance; set up GitHub repository; install libraries; create Claude brainstorm session	2–3 weeks	March 17, 2026	April 15, 2026	Bartimeus + Karikari
2 — Literature	Complete 10-paper review table; write synthesis; verify all DOIs; identify 3 baselines	2–3 weeks	March 17, 2026	March 31, 2026	Oti + Antwi
3 — Data	Download / collect dataset; clean; preprocess; generate and validate synthetic data if needed	2–4 weeks	March 17, 2026	March 31, 2026	Antwi
4 — Baseline	Implement 3 baseline models from related work; establish benchmark performance	1–2 weeks	March 27, 2026	April 3, 2026	Antwi
5 — Model	Implement primary Tier C model; tune hyperparameters; run experiments	3–4 weeks	April 1, 2026	April 15, 2026	Bartimeus + Oti
6 — Evaluation	Full evaluation: all planned metrics, visualisations, statistical significance tests	1–2 weeks	April 13, 2026	April 19, 2026	Manu
7 — Writing	Write full paper draft: all sections, in journal style	3–4 weeks	March 25, 2026	April 20, 2026	Bartimeus (Lead)
8 — Review	Supervisor review; revisions; proofreading; final formatting	1–2 weeks	April 21, 2026	April 24, 2026	All
9 — Submission	Submit to Target Journal 1	1 week	April 25, 2026	April 25, 2026	Bartimeus

SECTION 11 — REFERENCES

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SECTION 12 — RESEARCH TEAM

Role	Full Name	Student / Staff ID	Email	Contribution
Lead Researcher	Bartimeus, M. N. O.	3382422	manuelbartimeus@gmail.com	Coordination, Literature, Conclusion
Co-Researcher	Karikari, D. K.	3394522	davidkarikari01@gmail.com	Introduction, Problems, Scope
Co-Researcher	Manu, R. P.	3398922	richeal607@gmail.com	Results, Statistics, Ethics
Co-Researcher	Antwi, J. T.	3372922	antwijeffrey09@gmail.com	Data, Programming, Integration
Co-Researcher	Oti, W. B.	3419922	otiboakyewisdom99@gmail.com	Methodology, Background, Architecture
Supervisor	Osei, E. O.		eric.opoku.osei.rd@gmail.com	Oversight, review, approval
Co-Supervisor (if any)				

SECTION 13 — SUPERVISOR SIGN-OFF

This proposal must be reviewed and signed by the supervisor before ethics clearance application and data collection.

Field	Supervisor Entry
Supervisor Name	Dr. Eric Opoku Osei
Signature	
Date of Review	
Status	☐ Approved as submitted ☐ Approved with minor revisions ☐ Requires major revision ☐ Rejected — resubmit
Comments / Conditions	